

BLUE LOTUS RESEARCH INSTITUTE

PhD-Level Investment Due Diligence — Materials — Integrated Steel

Cleveland-Cliffs Inc.

[CLF] — Materials — Integrated Steel Due Diligence

US Steel Consolidator — \$6.9B Market Cap · Loss-Making Trough · Iron Ore Vertical Integration · EV
Auto Steel · Turnaround Candidate

August 2026 · Blue Lotus Research Institute · WO-DD-MEGA-MATH-20260814

CLASSIFICATION: CONFIDENTIAL — INSTITUTIONAL RESEARCH

SECTION 1 — EXECUTIVE SUMMARY

Verdict: Speculative Buy | Conviction: 6.8/10

Cleveland-Cliffs Inc. (CLF) — Materials — Integrated Steel PhD Due Diligence. All prices and financials from BL3 MySQL (bluelotus3.historical_prices, ticker_fundamentals). Zero Claude training-data figures. Verdict: Speculative Buy. Conviction: 6.8/10.

Metric	Value
Price (BL3, 2026-08-07)	\$12.26
Market Cap	\$6.93B
P/E (TTM)	N/A (loss-making)
P/B	1.24x
EPS (TTM)	N/A (-\$2.91 loss)
Net Income (TTM)	-\$1.66B (loss)
Dividend Yield	0.00%
52-Week High	\$16.70
52-Week Low	\$7.73

SECTION 2 — COMPANY OVERVIEW, BULL & BEAR CASE

Bull Case:

- Only US vertically integrated flat-rolled steel producer: Cleveland-Cliffs mines iron ore (Minnesota iron range), produces pellets, and operates integrated blast furnace/electric arc furnace steel mills — no other US producer controls the full iron ore-to-finished-steel supply chain; cost advantage in a tariff-protected US market
- Automotive steel dominance: CLF supplies 40%+ of the exposed automotive steel consumed in North America — advanced high-strength steel (AHSS) and stamped components for Ford, GM, and Stellantis; automotive replacement cycle and EV body-in-white steel demand supports volumes
- US steel tariffs Section 232 protection: 25% tariffs on imported steel make US-produced steel structurally competitive with imports — CLF's domestic production benefits from a protected market that prevents below-cost Chinese imports from competing directly
- Stelco acquisition (2024) adds low-cost EAF capacity: CLF acquired Canadian steelmaker Stelco, adding 3M+ tonnes of low-cost electric arc furnace capacity — EAF steel uses scrap (no coal dependency) and has structurally lower carbon emissions; positions CLF for green steel premium pricing
- Trough valuation: at 1.24x book value and deeply loss-making, CLF trades near liquidation value of its physical assets — the embedded option value in a steel upcycle is significant; automotive recovery and infrastructure spending could restore profitability rapidly

Bear Case:

- Currently loss-making: net income of -\$1.66B and EPS of -\$2.91 reflect the steel downcycle — input cost inflation (iron ore pellets, natural gas, scrap) while finished steel prices declined created a margin squeeze; recovery timeline is uncertain
- High debt load from acquisitions: CLF made multiple large acquisitions (AK Steel, ArcelorMittal USA, Stelco) — the combined leverage is significant relative to the current earnings trough; debt service constrains financial flexibility during low steel price environments

- US automotive production risk: CLF's largest end market is automotive — any UAW strike, EV ramp-up delays, or consumer credit tightening that reduces vehicle production directly impairs CLF steel demand by millions of tonnes
- Chinese steel export surge: China has historically dumped excess steel globally when domestic demand is weak — even with Section 232 tariffs, Chinese overcapacity creates global price pressure that reduces US steel prices and CLF margin
- EAF competition from NUE/STLD: Nucor and Steel Dynamics operate more cost-efficient electric arc furnaces that use scrap — CLF's blast furnace-heavy model carries higher fixed costs and energy intensity than EAF mini-mills

SECTION 3 — FINANCIAL PERFORMANCE (BL3 MySQL)

Financial Metric (BL3 MySQL)	Value
P/E (TTM)	N/A (loss-making)
Forward P/E	N/A (loss-making)
P/B	1.24x
Net Income (TTM)	-1.660B (loss)
EPS (TTM)	\$-2.91 (loss)
Market Cap	\$6.9B
Shares Outstanding	554M
Dividend Yield	0.00%
52-Week High	\$16.70
52-Week Low	\$7.73
Price (BL3, 2026-08-07)	\$12.26

SECTION 4 — PORTER'S FIVE FORCES (MATERIALS — INTEGRATED STEEL)

Force	Intensity	Analysis
Rivalry	MEDIUM-HIGH	Iron ore (VALE vs Rio Tinto/BHP), aluminium (AA vs Alcoa/Rusal/Chalco), and steel (CLF/NUE vs domestic mills) are intensely competitive on price; commodity businesses compete primarily on cost of production rather than product differentiation. Fertiliser (NTR/MOS) is a global oligopoly driven by nutrient scarcity and Canpotex export cartel (NTR/Mosaic/Nutrien control 30%+ of global potash). ADM competes with Cargill, Bunge, and Louis Dreyfus in a global grain processing oligopoly. CL (consumer staples) competes in a rational duopoly/oligopoly (Procter & Gamble, Unilever). OXY competes against majors (Exxon, Chevron) in US Permian — cost curves determine competitive position. SentinelOne (S) competes intensely with CrowdStrike, Microsoft Defender, Palo Alto in EDR/XDR.
New Entrants	LOW	Mining requires \$5-20B in capital, multi-decade permitting, and geological assets that cannot be created; no new US integrated steel mill has been built in 30+ years. Fertiliser mining (potash, phosphate) requires access to geological deposits that are geographically concentrated (Canada, Morocco, Russia, China). ADM's grain elevator and processing network took a century to build. Colgate's brand (120+ year trust in 200 countries) cannot be replicated. Oil & gas requires access to acreage, drilling expertise, and midstream infrastructure. Cybersecurity (SentinelOne) has lower capital barriers but requires AI model training data at scale — 8M+ endpoints provides an increasingly hard-to-replicate training advantage.
Supplier Power	MEDIUM	Mining companies control their own deposits (VALE owns

		Carajás, world's largest iron ore mine) — own supplier position. Steel mills (CLF, NUE) are buyers of scrap metal and iron ore; scrap prices are market-set with moderate supplier leverage. NTR controls Potash Corp legacy mines but nitrogen requires natural gas (Henry Hub price-driven). ADM buys grains from farmers at commodity prices — no single farmer has pricing power. OXY operates in the Permian where service company costs (Halliburton, SLB) carry moderate leverage during drilling activity peaks. SentinelOne buys cloud compute (AWS/Azure) — standard market pricing.
Buyer Power	MEDIUM	Steel buyers (automakers, construction) are sophisticated procurement operations that negotiate volumes annually — some switching between domestic mills (CLF, NUE, STLD) based on price. Agricultural buyers (grain traders, food processors) purchase ADM's output at commodity prices. Consumer goods buyers (supermarkets, mass market retailers) negotiate hard with CL — private label is the credible threat. Oil buyers (refiners) buy at spot — no single buyer commands pricing power. Cybersecurity buyers are large enterprises — budget cycles are multi-year but competitive bids occur at contract renewal; Microsoft's bundled Defender creates perpetual competitive pressure on SentinelOne.
Substitutes	MEDIUM	Aluminium substitutes steel in automotive (AA benefits); carbon fibre substitutes both at lower scale. Scrap steel recycling substitutes virgin iron ore in electric arc furnaces (NUE's model); VALE is exposed to DRI/HBI adoption. Potash has no substitute for crop nutrition — agricultural yield loss from potassium deficiency is immediate and severe. Alternative proteins (insects, lab meat) are very long-term ADM substitutes. Consumer staples

		substitutes are private label — Colgate has resisted private label in oral care more than most CPG peers. Natural gas substitutes coal for power (positive for OXY's production mix). AI-powered security substitutes or supplements third-party EDR — Microsoft Copilot for Security is the largest substitute risk for SentinelOne.
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SECTION 5 — KEY RISKS

- Commodity price cyclicalities: metals (VALE, AA, CLF, NUE), fertilisers (NTR, MOS), and oil (OXY) all experience multi-year price cycles driven by global supply-demand imbalances; earnings can collapse 60-80% from peak to trough in a single cycle
- China demand dependency: iron ore (VALE), aluminium (AA), potash (NTR), and phosphate (MOS) are all heavily dependent on Chinese agricultural and industrial demand — any Chinese economic slowdown creates immediate volume and price pressure
- Geopolitical supply disruption risk: Russia/Belarus potash sanctions, Middle East oil supply disruptions, and South American mining political risk all create potential supply-demand shocks in opposite directions across the portfolio
- Carbon transition and decarbonisation pressure: steel (CLF, NUE), fertiliser production, oil (OXY), and grain processing (ADM) are carbon-intensive industries facing increasing regulatory pressure, carbon taxes, and investor ESG mandates
- Cybersecurity competition (SentinelOne): Microsoft's bundled Defender for Endpoint, CrowdStrike's market recovery, and Palo Alto's XSIAM platform all represent credible competitive threats; the AI security market is evolving rapidly and the winner is not yet determined
- ADM accounting investigation: ongoing SEC investigation into ADM's nutrition segment intersegment transactions creates headline risk and potential earnings restatement; management credibility has been partially impaired

SECTION 6 — INVESTMENT THESIS & VERDICT

VERDICT: SPECULATIVE BUY | CONVICTION: 6.8/10

Cleveland-Cliffs Inc. (CLF) — Speculative Buy with 6.8/10 conviction. BL3 data: Price \$12.26, MCap \$6.9B. Materials, energy, and cybersecurity represent essential industrial infrastructure, food security, and digital security — each with distinct moats and cyclical characteristics.

Data: BL3 MySQL bluelotus3.ticker_fundamentals + historical_prices | CivilisationOS RAG (WSJ/FT/NIKKEI). Work Order: WO-DD-MEGA-MATH-20260814. Report Date: 14 August 2026. Zero training-data figures. AMP compliant.